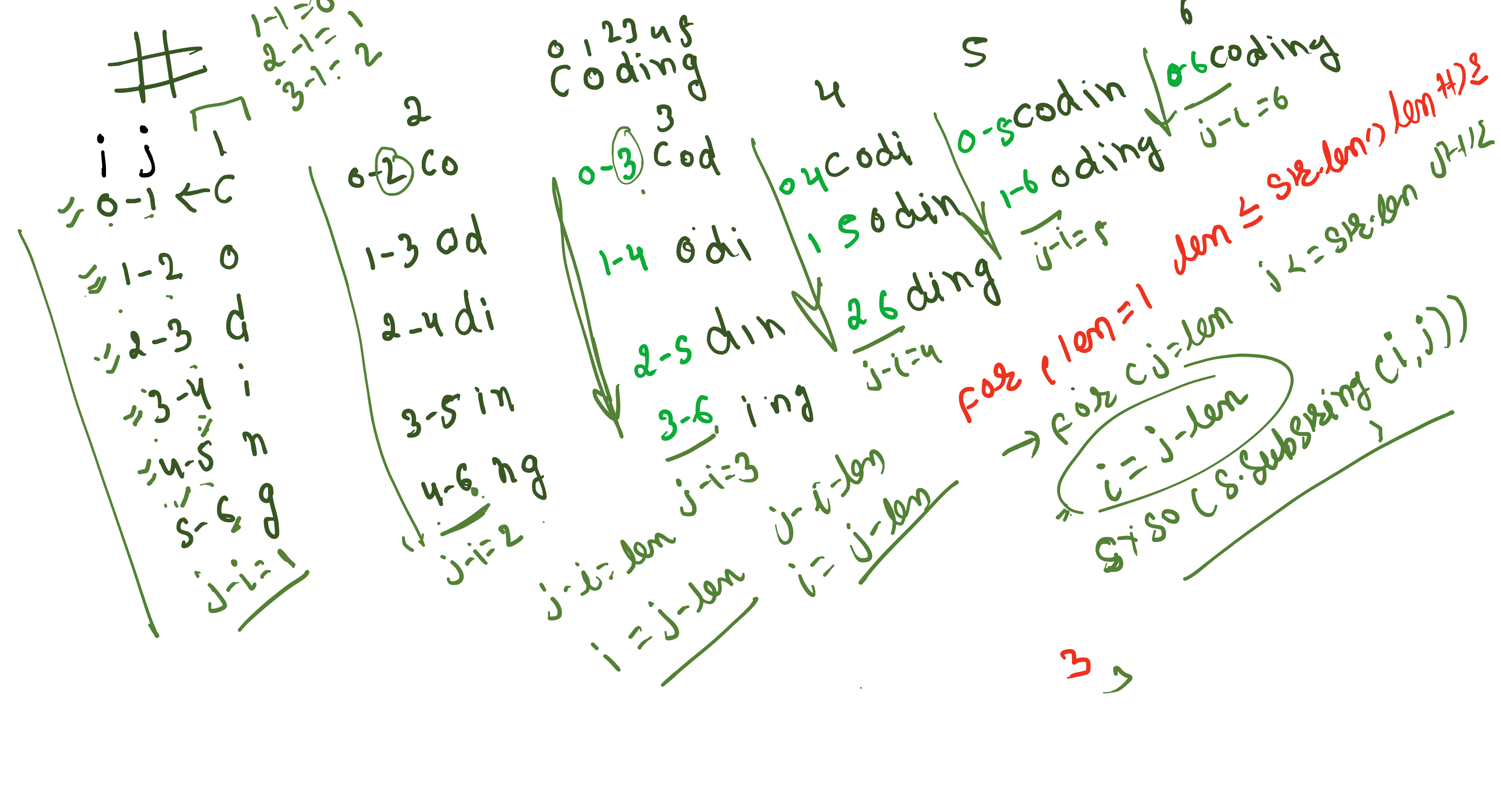
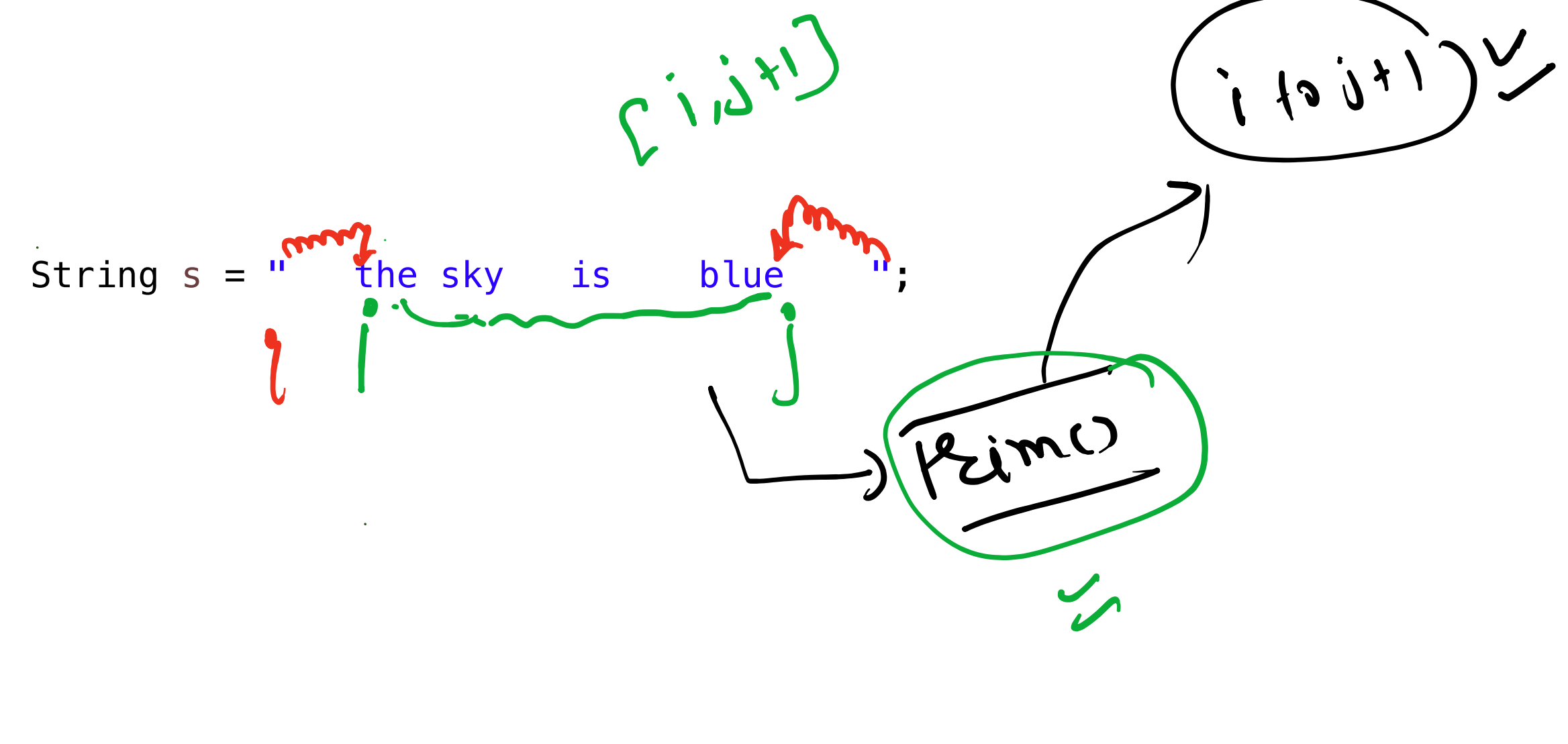




Kartik Bhaiya and Kanak Bhaiya are discussing a special type of number, which they call a **Coding Blocks Number (CB Number)**. A number qualifies as a CB Number based on the following criteria:

- 0 and 1 are not considered CB numbers.
- The following prime numbers are always CB numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
- Any other number is a CB number if it is not divisible by any of the numbers listed in point (2).

Handwritten notes include "31, 37", "1143", and "not divisible".

011 x CB X

The Challenge

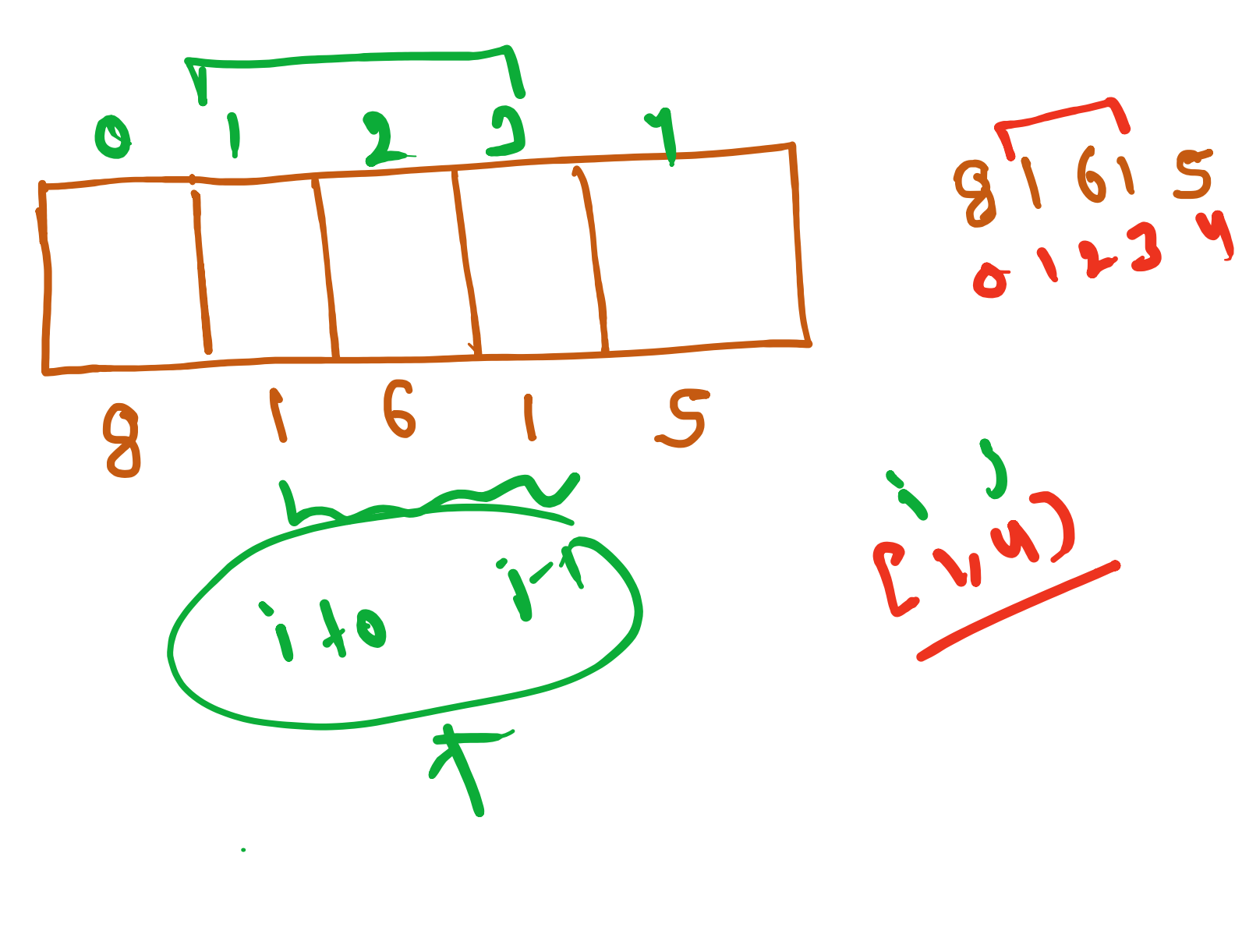
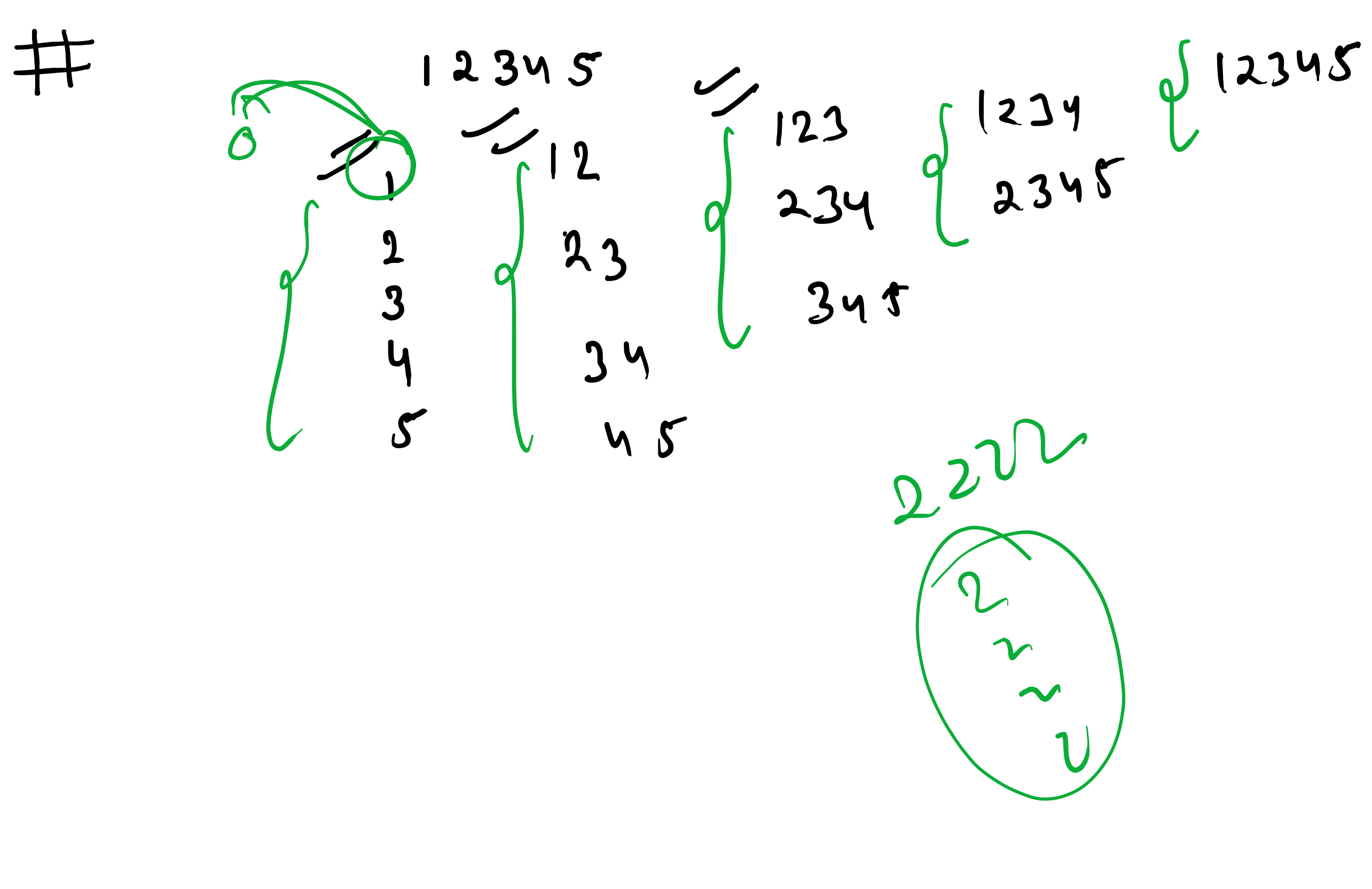
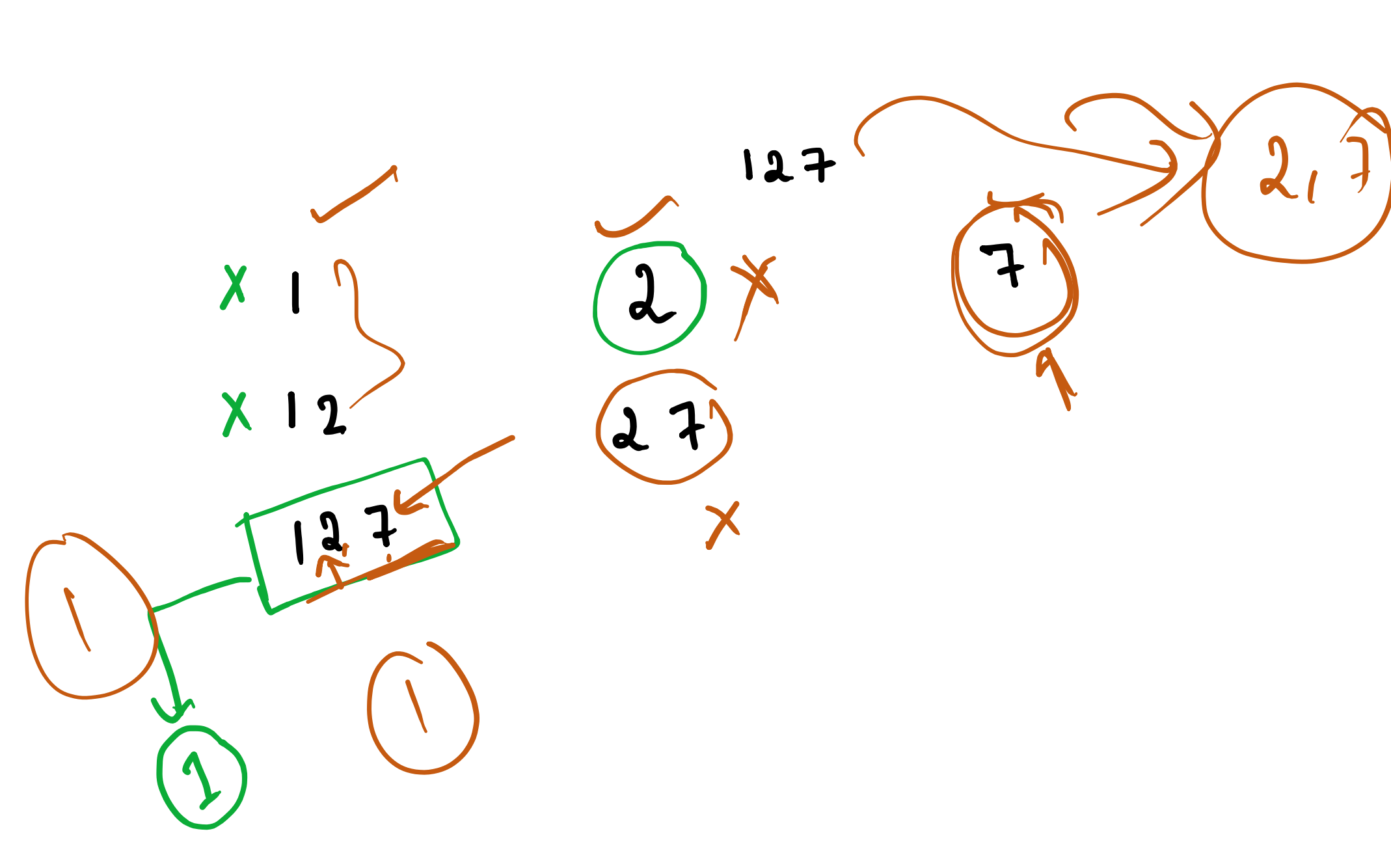
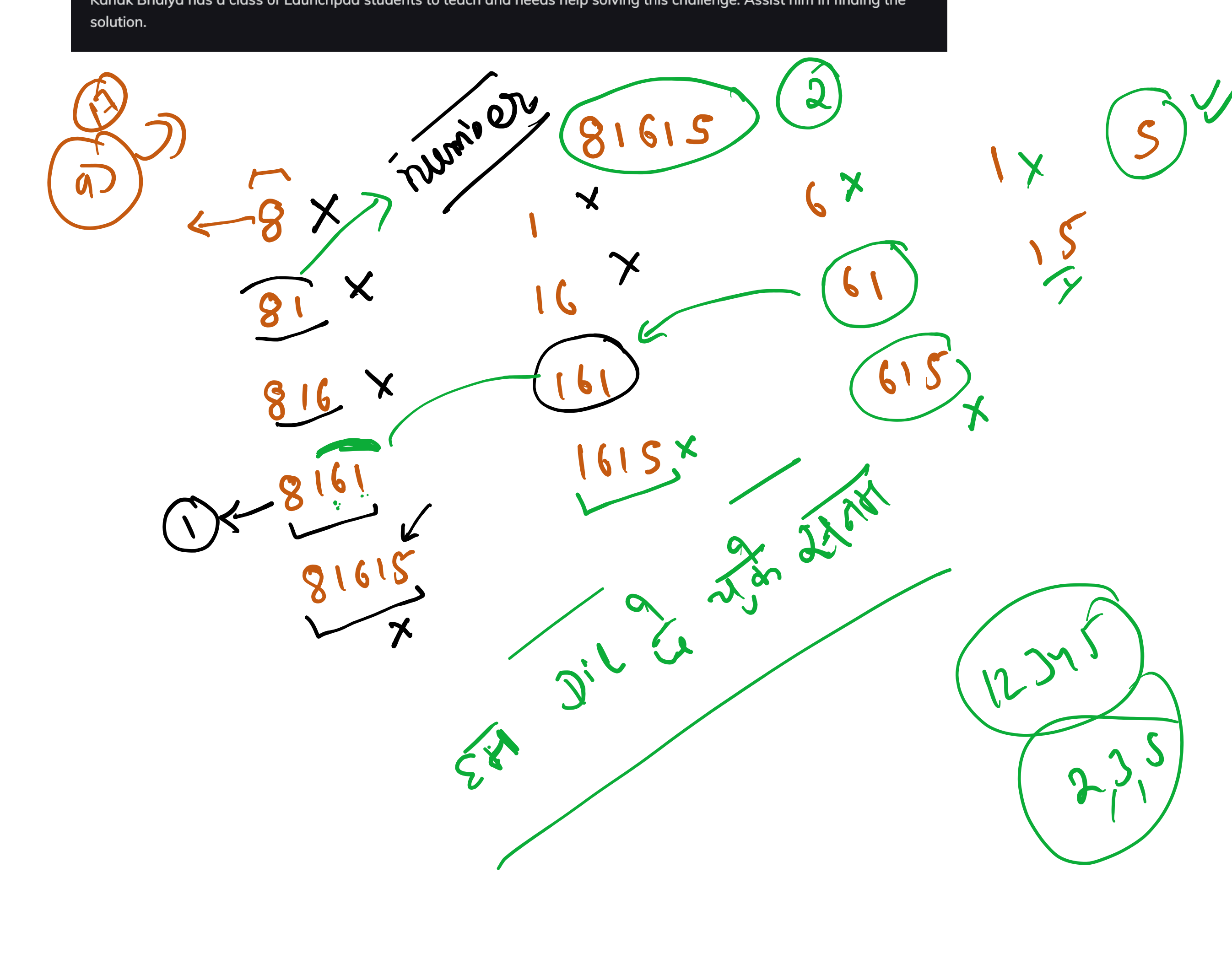
Kartik Bhaiya challenges Kanak Bhaiya with a problem:

He provides a string of digits, and Kanak Bhaiya must determine the maximum number of CB numbers that can be extracted from it while following these constraints:

- Non-overlapping Substrings:
 - A CB number cannot be a substring or superstring of another chosen CB number.
 - Example: In 4991, both 499 and 991 are CB numbers, but we can choose only one of them.
- Valid Substring Selection:
 - The CB number must be a contiguous substring of the given string.
 - Example: In 481, we cannot select 41 as a CB number because 41 is not a contiguous substring of "481".
- Maximization Goal:
 - Since multiple solutions may exist, the goal is to find the maximum number of CB numbers that can be extracted from the given string.

Kanak Bhaiya has a class of Launchpad students to teach and needs help solving this challenge. Assist him in finding the solution.

81615 → CB Number



```
public static int count_cb_number(String s) {
    int c = 0;
    boolean[] visited = new boolean[s.length()];
    for (int len = 1; len <= s.length(); len++) {
        for (int j = len; j <= s.length(); j++) {
            int i = j - len;
            long num = Long.parseLong(s.substring(i, j)); // i to j-1
            if (isCbNumber(num) && !visited(i, j - 1)) {
                // marked
                for (int k = i; k < j; k++) {
                    visited[k] = true;
                }
                c++;
            }
        }
    }
    return c;
}

public static boolean isnotvisited(boolean[] visited, int si, int ei) {
    // TODO Auto-generated method stub
    for (int i = si; i <= ei; i++) {
        if (visited[i]) {
            return false;
        }
    }
    return true;
}
```

Handwritten notes include "len=3", "len=1", "j=1", "j=0", and a diagram showing a sequence of digits "0 1 2" with "x" and "1 2 7" below it.

